Tontadarya College of Engineering, Gadag

ETHNOTECH ACADEMIC SOLUTIONS

DEPARTMENT ELECTRONICS AND COMMUNICATION

Report on:

Smart Food Wastage Management System

Report submitted for the award of

Assignment marks of Subject Name: Python

Full Stack

6th Semester (2025-26)

Submitted by:

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Submitted To:
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Technical Trainer
Ethnotech Academy

➢ ABSTRACT

This project focuses on developing a Smart Food Waste Management System to reduce food wastage in educational institutions such as canteens, hostels, and mess facilities. These institutions often generate significant food waste due to over-preparation, lack of demand forecasting, and inefficient inventory management.

The system introduces a digital platform that enables real-time tracking, predictive planning, and coordination among stakeholders. It allows students to pre-register meal preferences, helping canteens prepare accurate quantities of food. Additionally, the system supports food waste tracking, inventory management using QR/barcode scanning, and surplus food donation to NGOs.

The application aims to improve operational efficiency, reduce environmental impact, and promote sustainable food practices.

Food wastage is a major issue in institutional environments where meals are prepared in bulk. Due to unpredictable attendance and lack of proper tracking systems, large quantities of food are wasted daily.

This Smart Food Waste Management System is designed to address these issues by introducing a centralized digital solution. The system helps in monitoring food consumption, predicting demand, and managing surplus food effectively.

It also encourages responsible consumption and supports sustainability initiatives within campuses.

PROBLEM STATEMENT

Educational institutions face challenges such as:

- Over-preparation of food
- Lack of accurate demand forecasting
- Manual tracking leading to errors
- No system for managing surplus food
- Inefficient inventory and procurement

Students and staff also lack a platform to pre-book meals or provide preferences, resulting in both shortages and wastage.

➢ OBJECTIVES OF THE PROJECT

- To reduce food wastage in institutions
- To provide meal pre-registration for students
- To implement real-time inventory tracking
- To monitor and analyze food waste patterns
- To enable surplus food donation to NGOs
- To provide analytics for better decision-making

SCOPE OF THE PROJECT

The system includes:

- Web/mobile-based interface
- Meal preference registration
- Inventory management using QR/barcode
- Waste tracking with image upload
- Donation system for excess food
- Admin dashboard for analytics

The system can be extended with mobile apps, AI-based prediction, and GPS tracking in future.

➢ LITERATURE SURVEY

Existing systems mainly focus on food delivery rather than waste management. Some institutions rely on manual logs, which are inefficient and inaccurate.

Modern smart systems emphasize:

- Data-driven decision making
- Real-time monitoring
- Automation in inventory and procurement

This project combines these features into a single platform tailored for institutional use.

➢ SYSTEM REQUIREMENTS

Hardware Requirements

- Laptop/Desktop
- Minimum 8GB RAM
- Internet Connection

Software Requirements

- Operating System: Windows/Linux/macOS
- Python (Django Framework)
- MySQL Database
- VS Code
- Web Browser

TECHNOLOGY STACK

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Django (Python)
- **Database:** MySQL
- **Additional Tools:** QR/Barcode Scanner Integration

➢ SYSTEM ARCHITECTURE

The system follows a centralized architecture:

- Users interact through web/mobile interface
- Backend processes data using Django
- Database stores food, user, and waste records
- Admin dashboard provides analytics and reports

➢ MODULE DESCRIPTION

1. Student Module

- Pre-register meal preferences
- View meal availability
- Track attendance

2. Canteen/Staff Module

- Manage food preparation
- Update inventory
- View demand forecasts

3. Waste Management Module

- Log food waste
- Upload images
- Categorize (edible/inedible)
- Track waste trends

4. Donation Module

- Share surplus food
- Connect with NGOs
- Schedule pickups

5. Admin Module

- View analytics dashboard
- Monitor waste metrics
- Generate reports
- Optimize procurement

➢ RESULT AND OUTPUT

The Smart Food Waste Management System was successfully developed and tested using the Django framework with a MySQL database. The system provides a user-friendly interface and efficient backend processing for managing food waste and donations.

ADVANTAGES

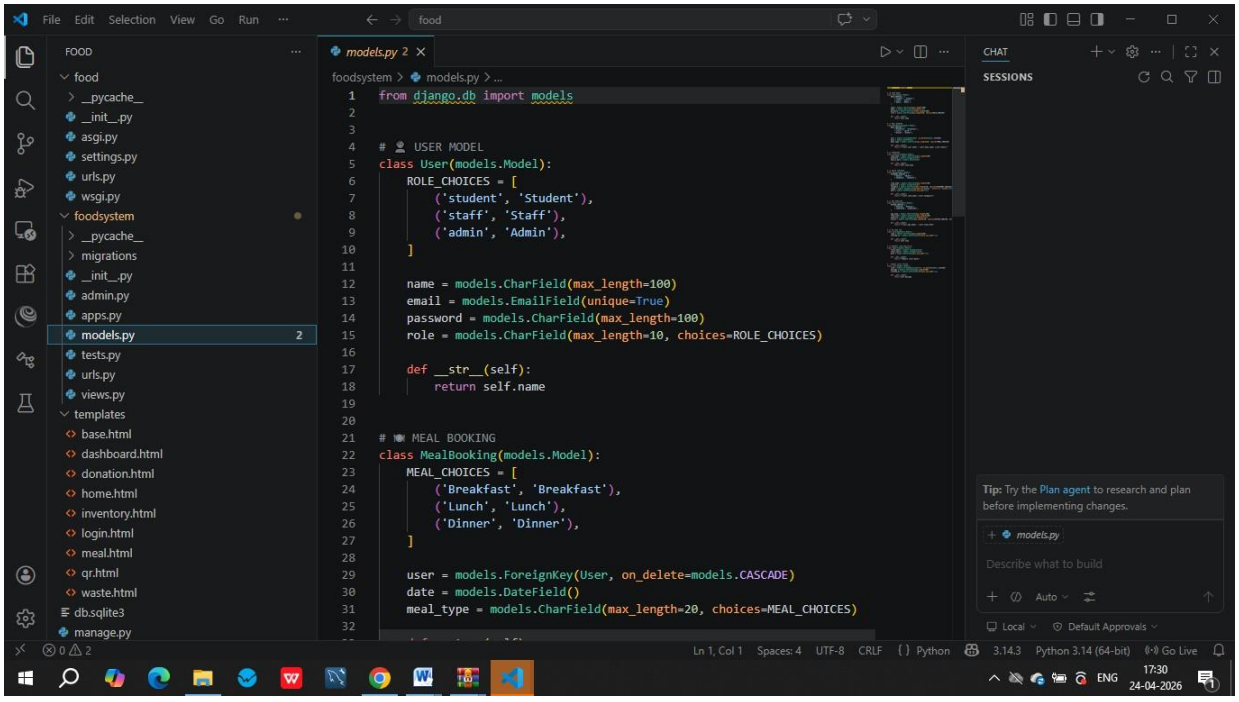
- Reduces food wastage significantly
- Improves resource utilization
- Supports environmental sustainability
- Provides real-time insights
- Encourages responsible consumption

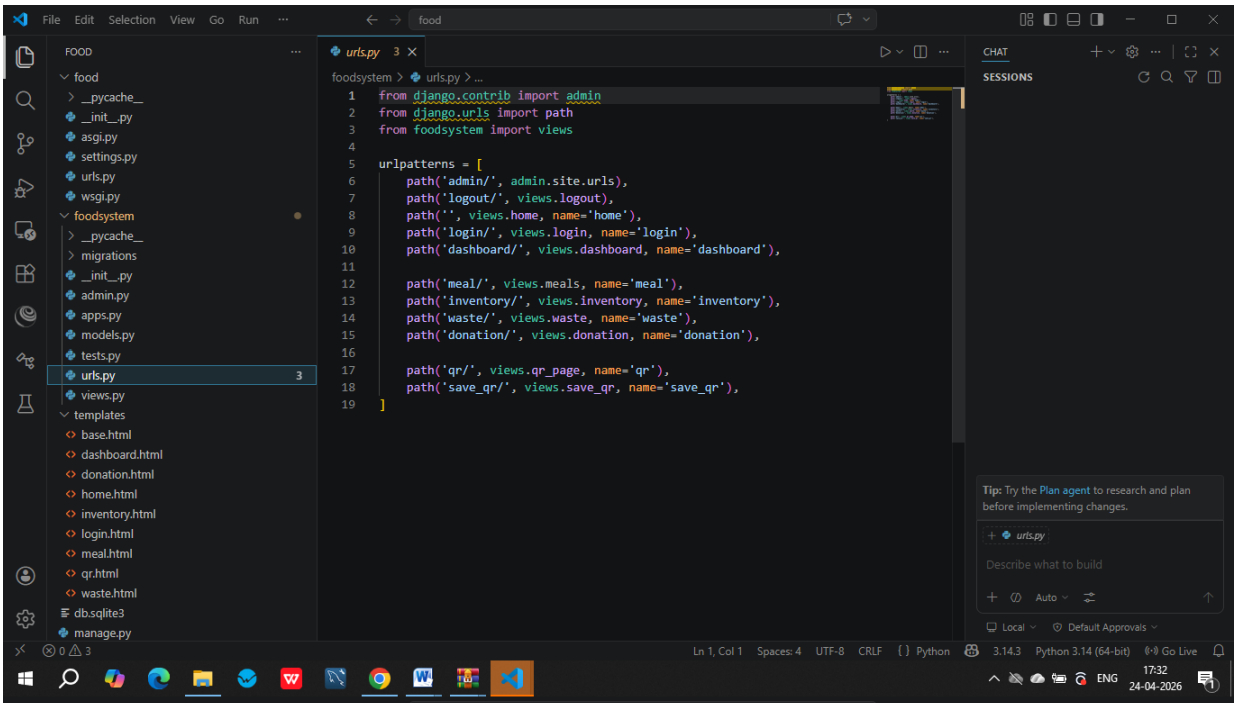
LIMITATIONS

- Requires internet connectivity
- Initial setup cost for institutions
- Depends on user participation

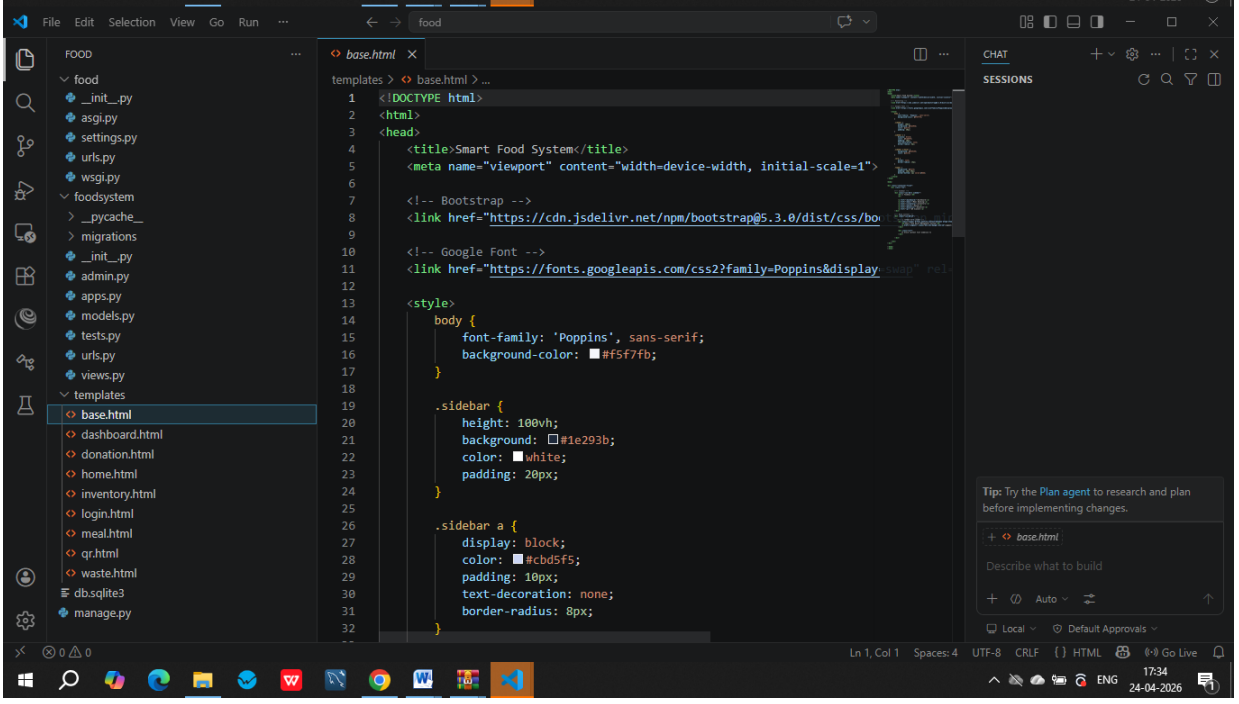
FUTURE ENHANCEMENTS

- AI-based demand prediction
- Mobile app development
- GPS-based food donation tracking
- Integration with government/NGO systems
- Automated notifications and alerts

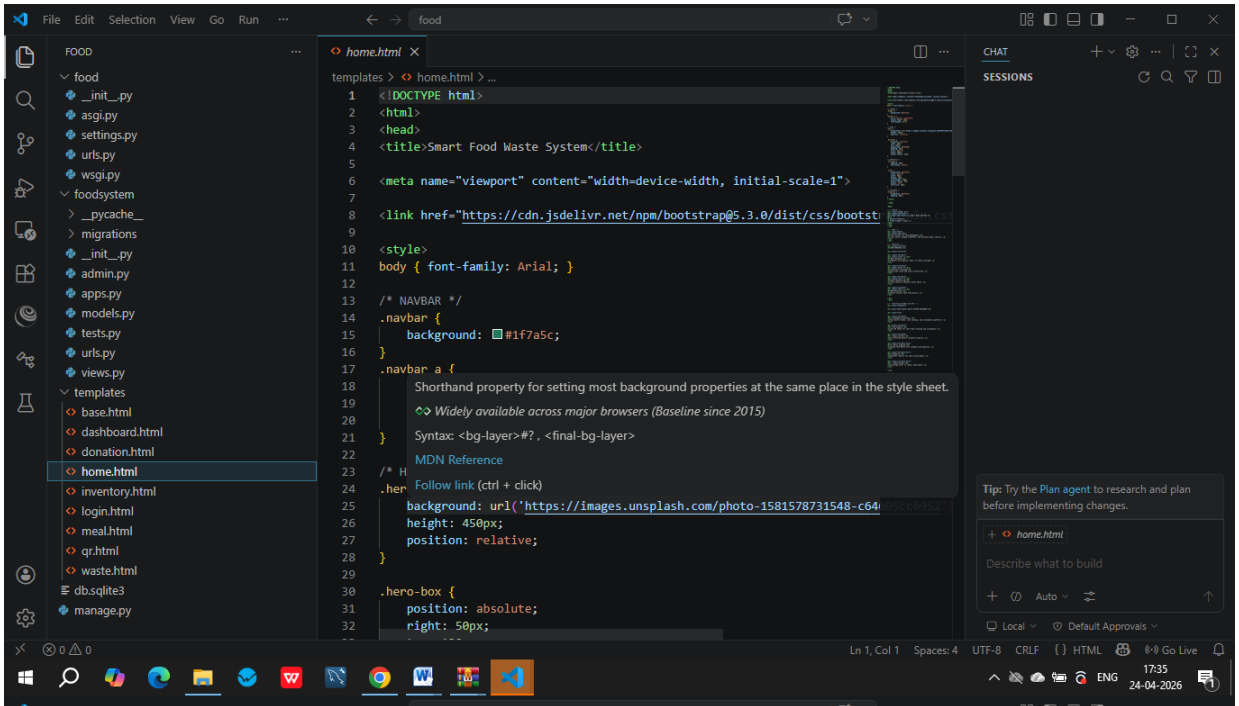


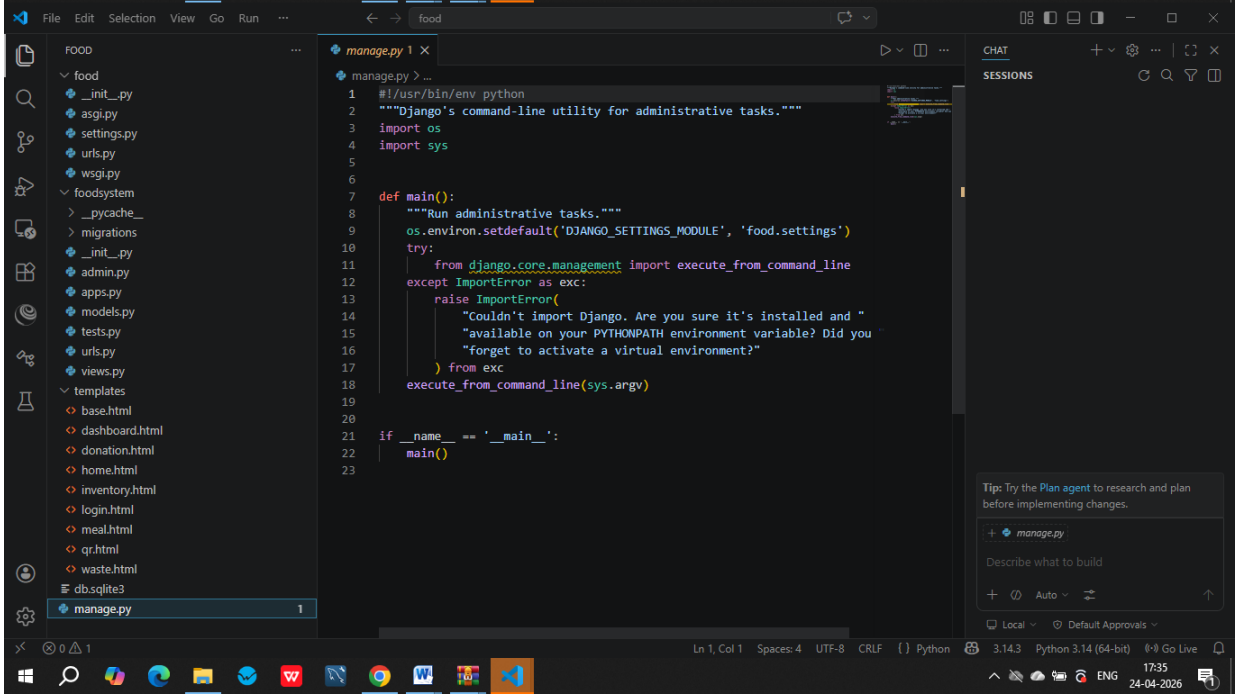


```python
from django.contrib import admin
from django.urls import path
from foodsystem import views

urlpatterns = [
    path('admin/', admin.site.urls),
    path('logout/', views.logout),
    path('', views.home, name='home'),
    path('login/', views.login, name='login'),
    path('dashboard/', views.dashboard, name='dashboard'),

    path('meal/', views.meals, name='meal'),
    path('inventory/', views.inventory, name='inventory'),
    path('waste/', views.waste, name='waste'),
    path('donation/', views.donation, name='donation'),

    path('qr/', views.qr_page, name='qr'),
    path('save_qr/', views.save_qr, name='save_qr'),
]
```



```html
<!DOCTYPE html>
<html>
<head>
    <title>Smart Food System</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">

    <!-- Bootstrap -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bo

    <!-- Google Font -->
    <link href="https://fonts.googleapis.com/css2?family=Poppins&display=swap" rel

    <style>
        body {
            font-family: 'Poppins', sans-serif;
            background-color: #f5f7fb;
        }

        .sidebar {
            height: 100vh;
            background: #1e293b;
            color: white;
            padding: 20px;
        }

        .sidebar a {
            display: block;
            color: #cbd5f5;
            padding: 10px;
            text-decoration: none;
            border-radius: 8px;
        }
```


File Edit Selection View Go Run
food
FOOD
food
__init__.py
asgi.py
settings.py
urls.py
wsgi.py
foodsystem
__pycache__
migrations
__init__.py
admin.py
apps.py
models.py
tests.py
urls.py
views.py
templates
base.html
dashboard.html
donation.html
home.html
inventory.html
login.html
meal.html
qr.html
waste.html
db.sqlite3
manage.py
home.html
templates > home.html > ...
<!DOCTYPE html>
<html>
<head>
<title>Smart Food Waste System</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootst
<style>
body { font-family: Arial; }
/* NAVBAR */
.navbar {
background: #1f7a5c;
}
.navbar a {
Shorthand property for setting most background properties at the same place in the style sheet.
Widely available across major browsers (Baseline since 2015)
Syntax: <bg-layer>#? , <final-bg-layer>
MDN Reference
Follow link (ctrl + click)
background: url('https://images.unsplash.com/photo-1581578731548-c64
height: 450px;
position: relative;
}
.hero-box {
position: absolute;
right: 50px;
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
home.html
Describe what to build
Auto
Local
Default Approvals
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
HTML
Go Live
ENG
17:35
24-04-2026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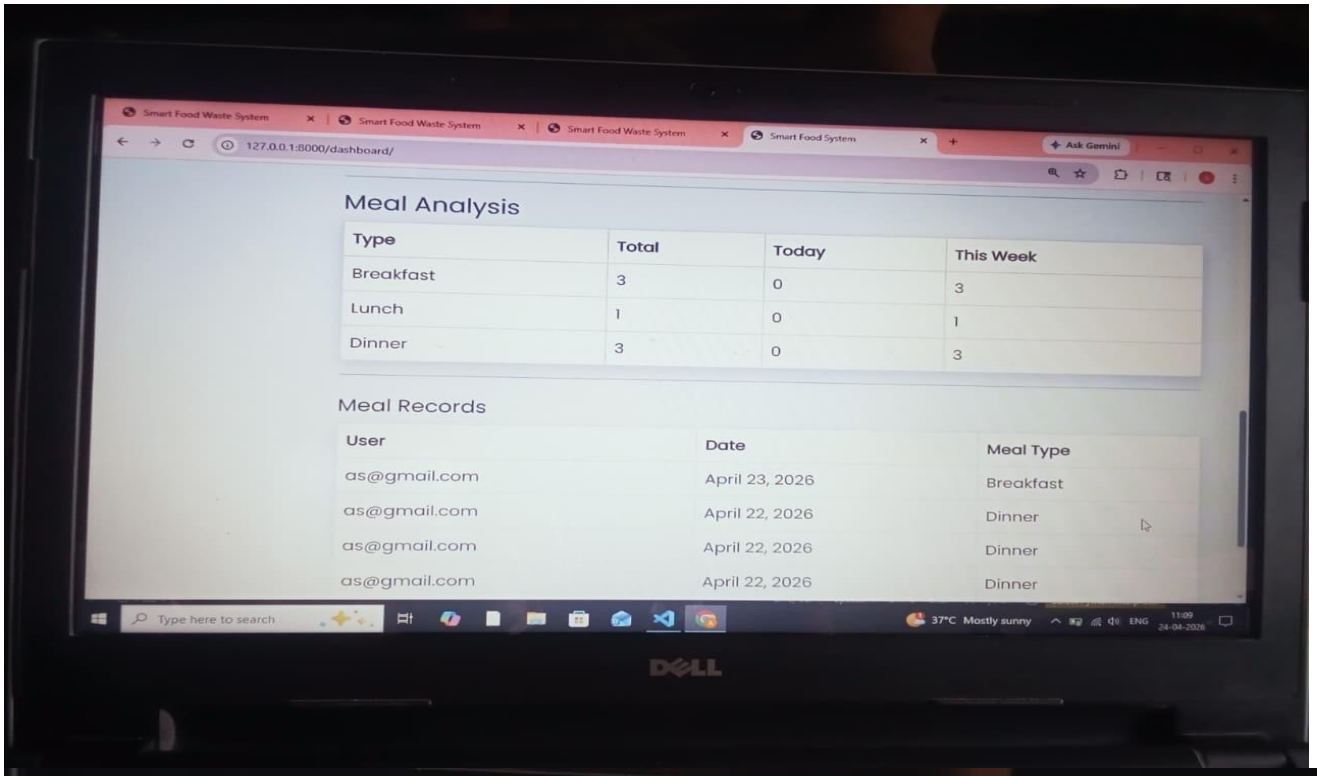
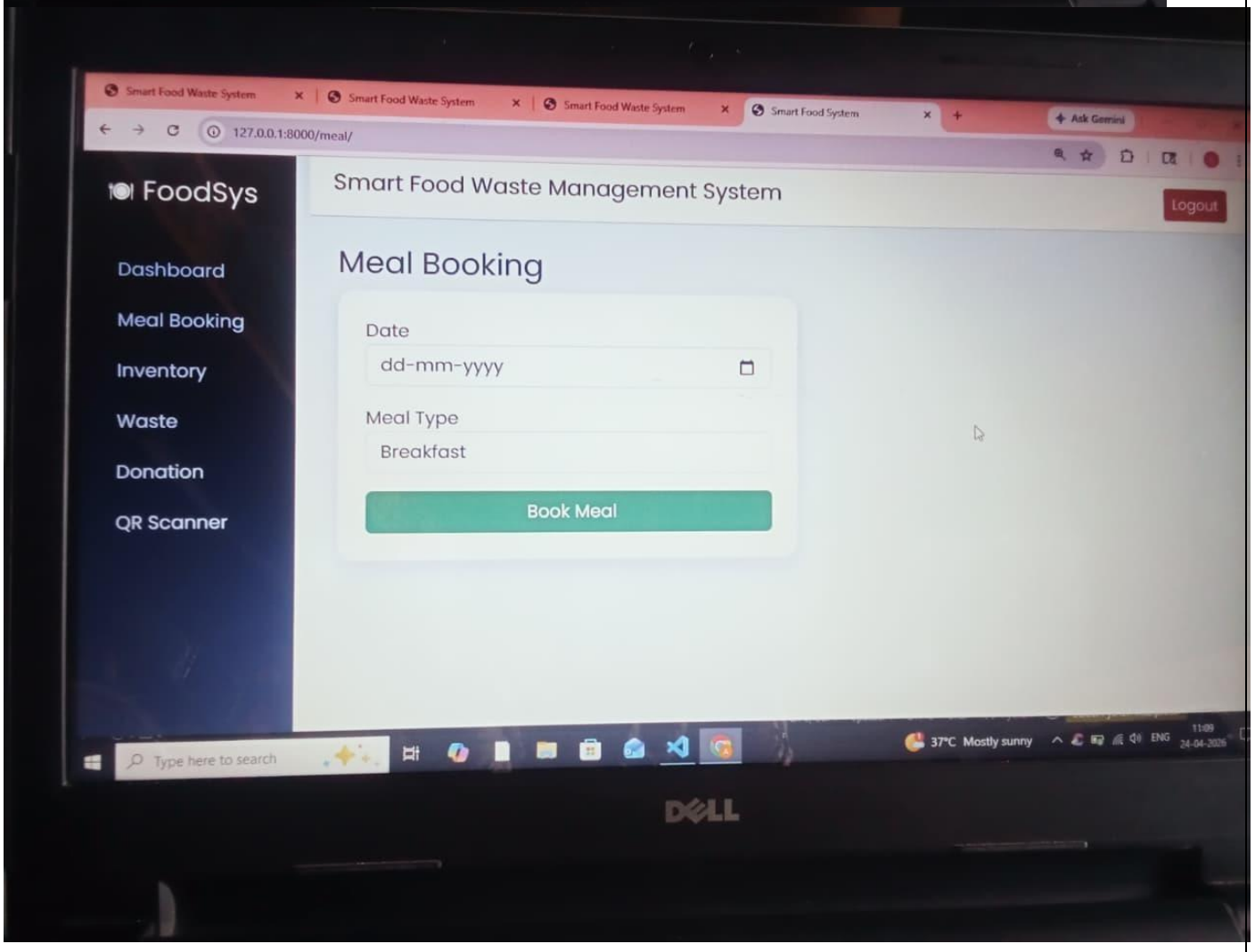
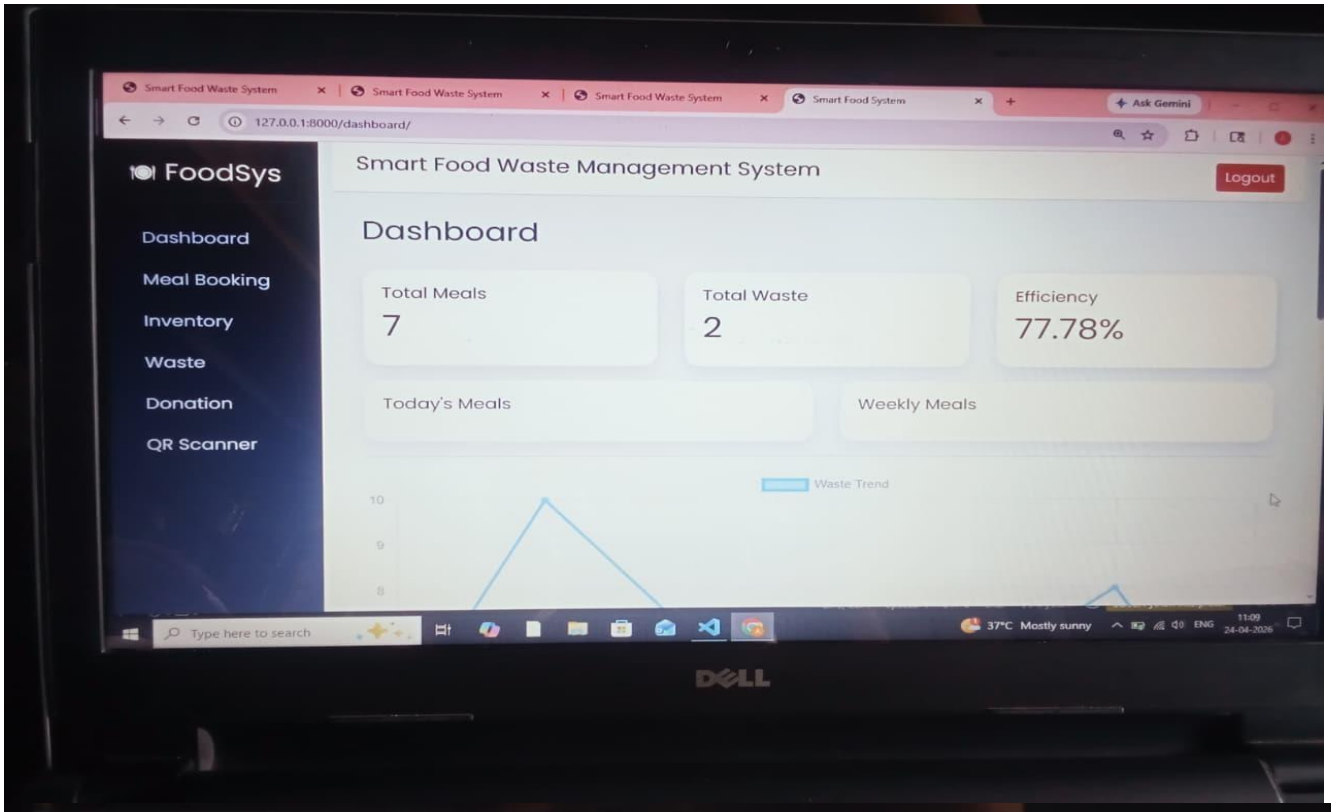
views.py
templates
base.html
dashboard.html
donation.html
home.html
inventory.html
login.html
meal.html
qr.html
waste.html
db.sqlite3
manage.py
manage.py 1
manage.py > ...
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys
def main():
"""Run administrative tasks."""
os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'food.settings')
try:
from django.core.management import execute_from_command_line
except ImportError as exc:
raise ImportError(
"Couldn't import Django. Are you sure it's installed and "
"available on your PYTHONPATH environment variable? Did you "
"forget to activate a virtual environment?"
) from exc
execute_from_command_line(sys.argv)
if __name__ == '__main__':
main()
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
manage.py
Describe what to build
Auto
Local
Default Approvals
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Python
3.14.3
Python 3.14 (64-bit)
Go Live
ENG
17:35
24-04-2026

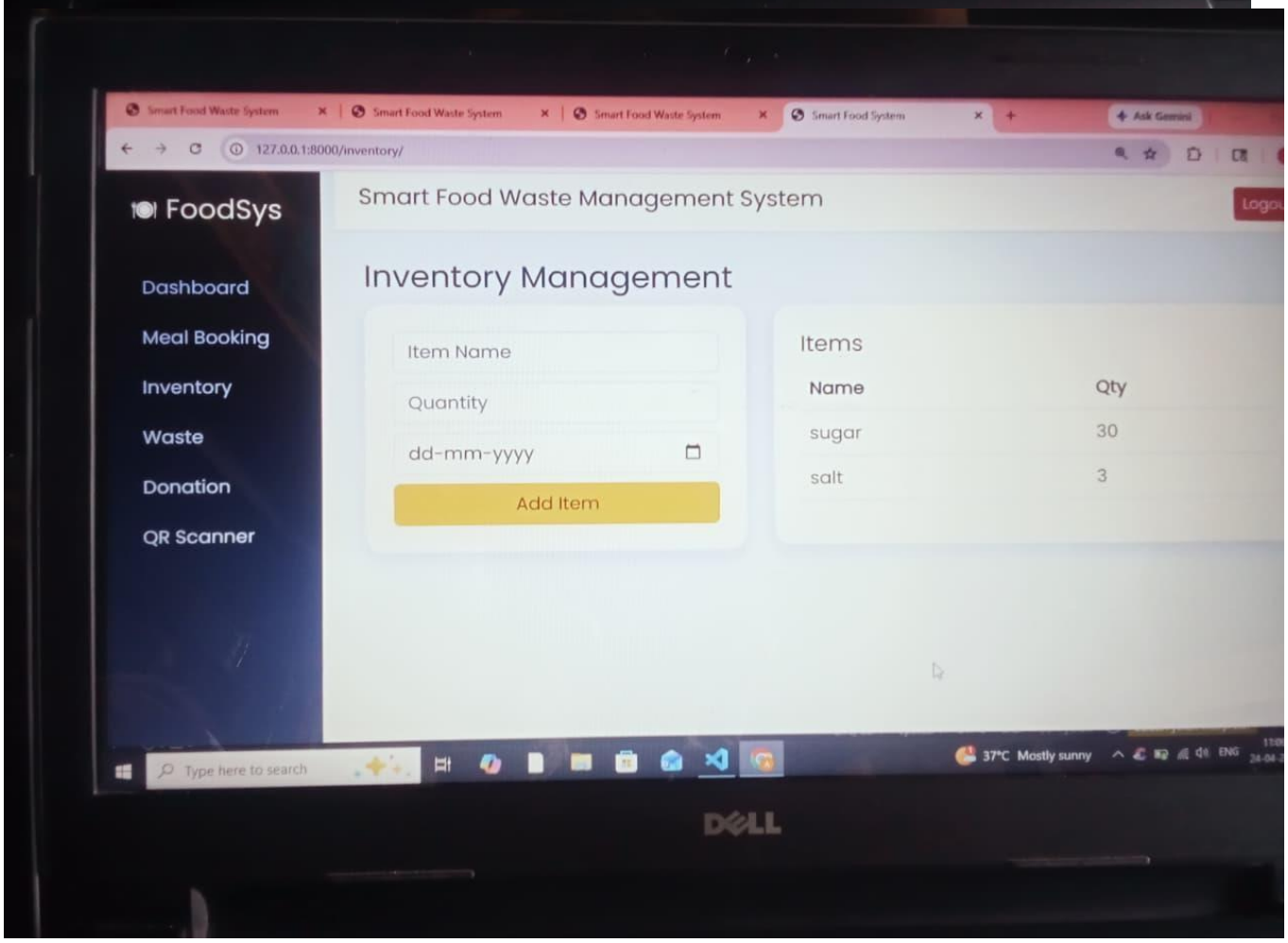
```python
from django.shortcuts import render, redirect
from django.http import JsonResponse
from .models import *
import json
from datetime import date, timedelta


# 🏠 HOME PAGE
def home(request):
    return render(request, 'home.html')


# 🔐 LOGIN
def login(request):
    if request.method == "POST":
        email = request.POST.get('email')
        password = request.POST.get('password')

        user = User.objects.filter(email=email, password=password).first

        if user:
            request.session['user_id'] = user.id
            request.session['role'] = user.role

            return redirect('/dashboard/')   # ✅ go to dashboard

        else:
            return render(request, 'login.html', {'error': 'Invalid credentials'})

    return render(request, 'login.html')
```

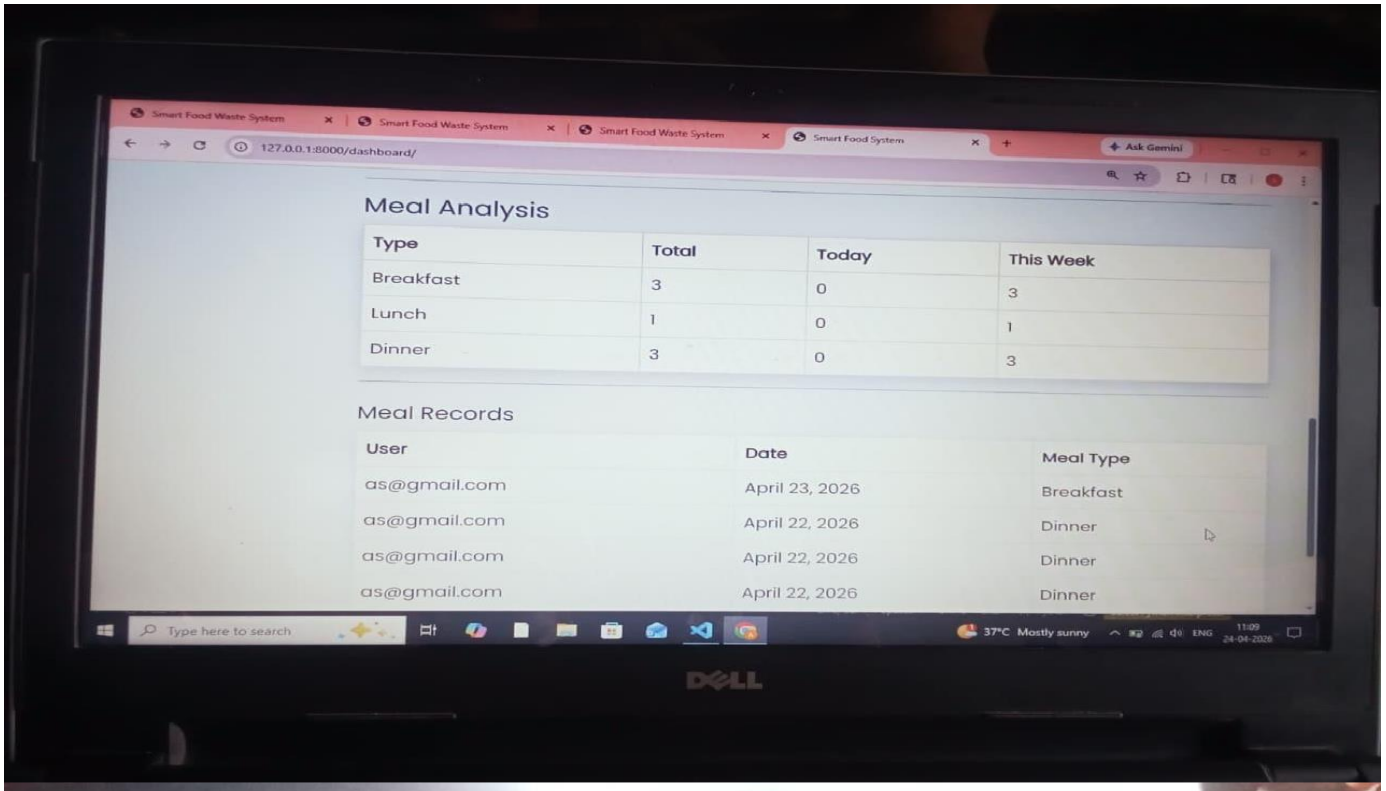
```python
# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY = 'django-insecure-=#*g45p11q$(r+_0s&=o#b*u-6t2&+-h1&m-b+!2jk

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = []


# Application definition

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'foodsystem',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'food.urls'
```


Smart Food Waste System
127.0.0.1:8000/dashboard/
Meal Analysis
Type | Total | Today | This Week
Breakfast | 3 | 0 | 3
Lunch | 1 | 0 | 1
Dinner | 3 | 0 | 3
Meal Records
User | Date | Meal Type
as@gmail.com | April 23, 2026 | Breakfast
as@gmail.com | April 22, 2026 | Dinner
as@gmail.com | April 22, 2026 | Dinner
as@gmail.com | April 22, 2026 | Dinner


Smart Food Waste System
127.0.0.1:8000/meal/
FoodSys
Smart Food Waste Management System
Logout
Dashboard
Meal Booking
Inventory
Waste
Donation
QR Scanner
Meal Booking
Date
dd-mm-yyyy
Meal Type
Breakfast
Book Meal


127.0.0.1:8000/dashboard/
FoodSys
Smart Food Waste Management System
Logout
Dashboard
Meal Booking
Inventory
Waste
Donation
QR Scanner
Dashboard
Total Meals
7
Total Waste
2
Efficiency
77.78%
Today's Meals
Weekly Meals
Waste Trend


127.0.0.1:8000/inventory/
FoodSys
Smart Food Waste Management System
Dashboard
Meal Booking
Inventory
Waste
Donation
QR Scanner
Inventory Management
Item Name
Quantity
dd-mm-yyyy
Add Item
Items
Name
Qty
sugar
30
salt
3


Smart Food Waste System
127.0.0.1:8000/dashboard/
Meal Analysis
Type | Total | Today | This Week
Breakfast | 3 | 0 | 3
Lunch | 1 | 0 | 1
Dinner | 3 | 0 | 3
Meal Records
User | Date | Meal Type
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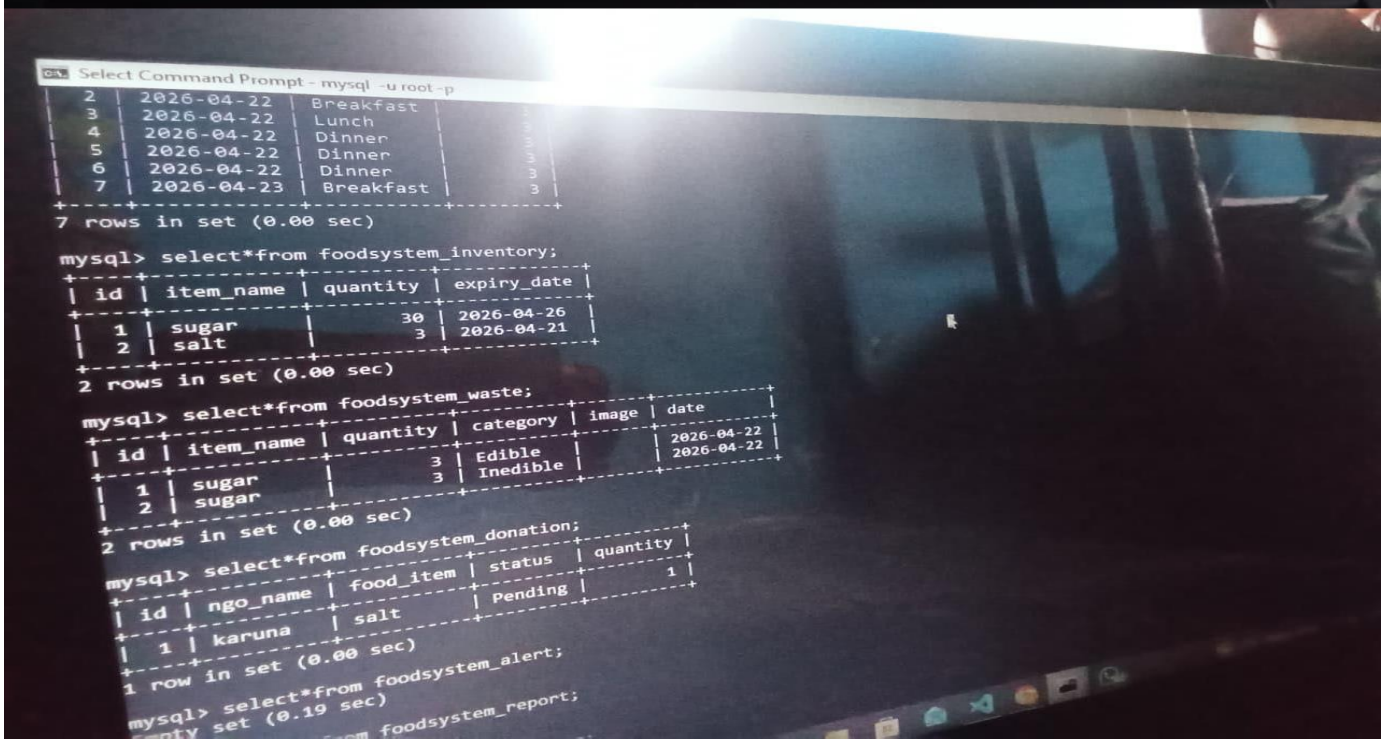
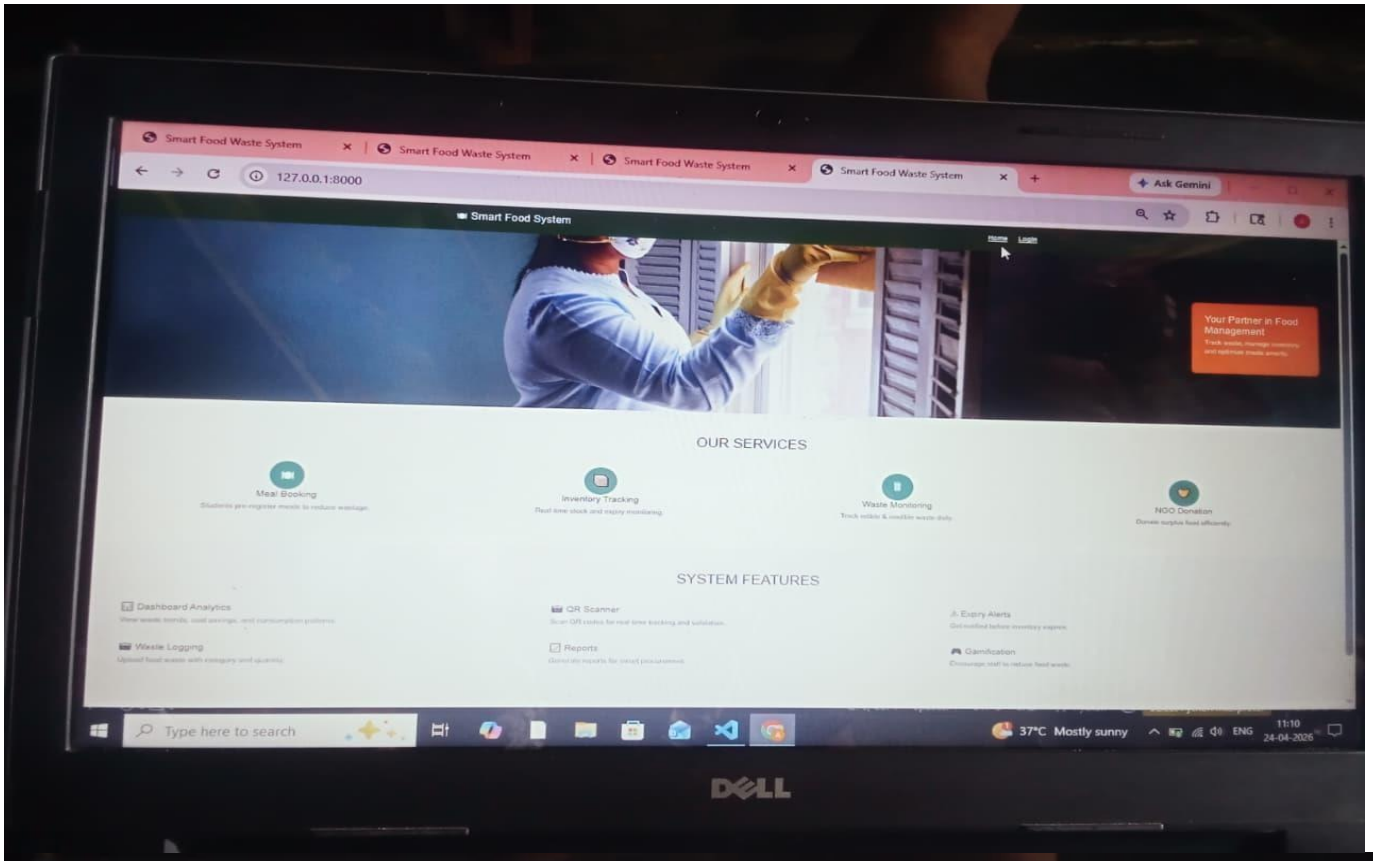
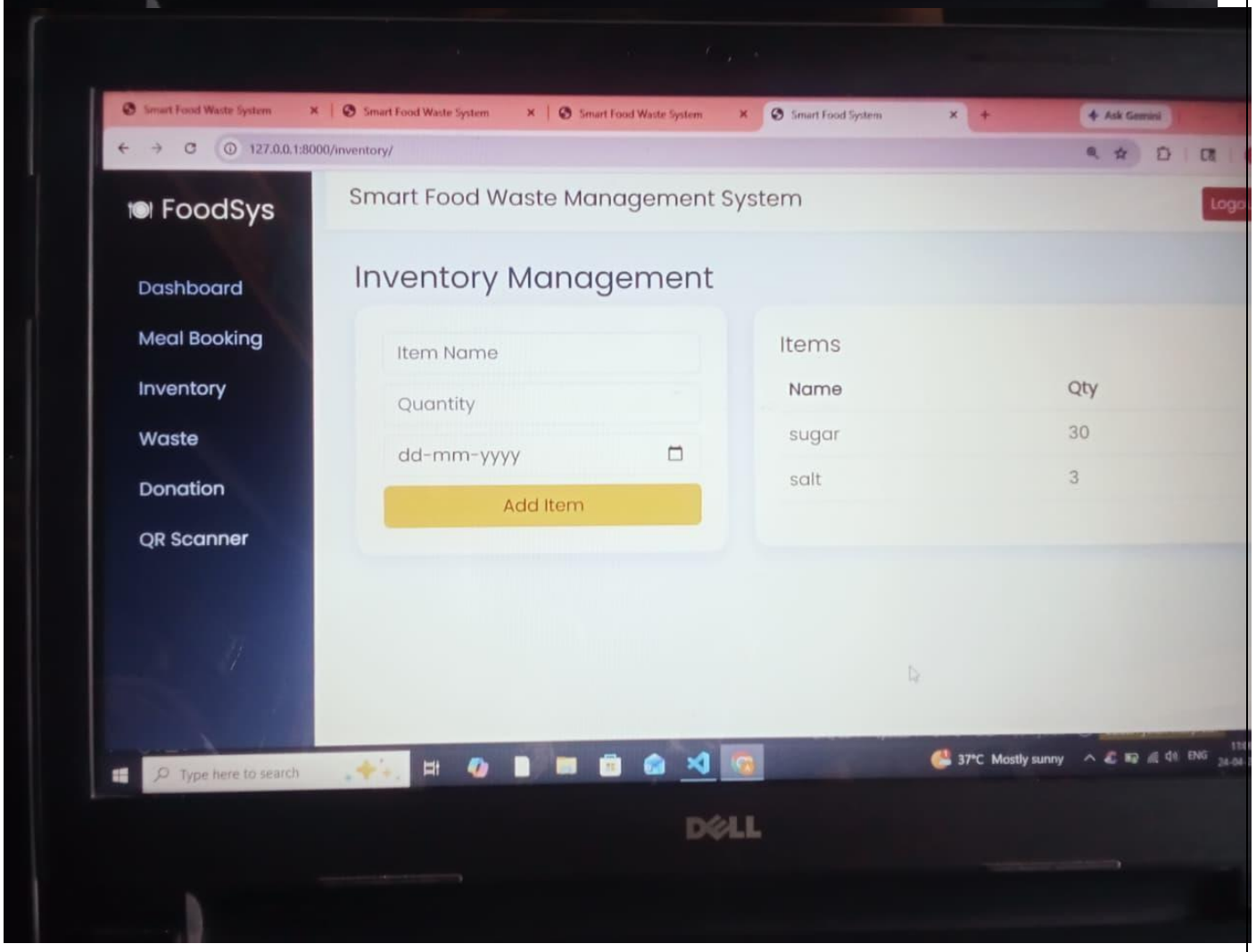
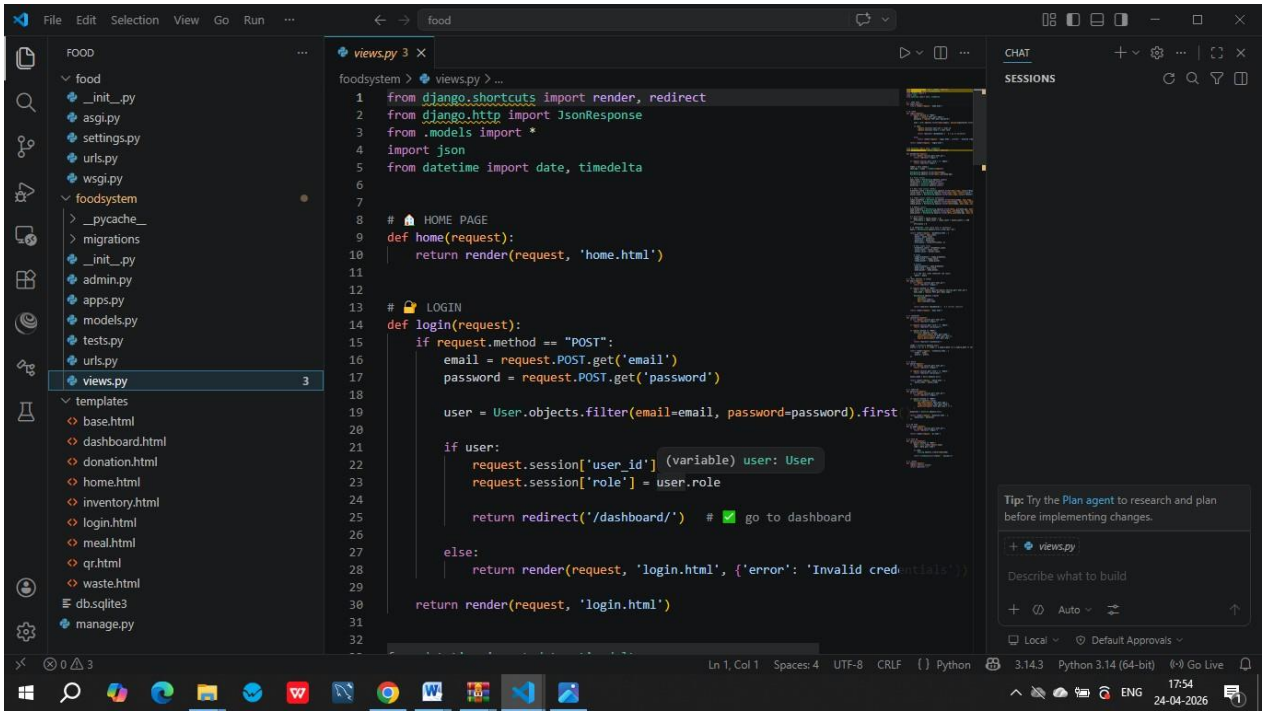
as@gmail.com | April 22, 2026 | Dinner
as@gmail.com | April 22, 2026 | Dinner
as@gmail.com | April 22, 2026 | Dinner


Select Command Prompt - mysql -u root -p
2	2026-04-22	Breakfast	3
3	2026-04-22	Lunch	3
4	2026-04-22	Dinner	3
5	2026-04-22	Dinner	3
6	2026-04-22	Dinner	3
7	2026-04-23	Breakfast	3
7 rows in set (0.00 sec)
mysql> select*from foodsystem_inventory;
id	item_name	quantity	expiry_date
1	sugar	30	2026-04-26
2	salt	3	2026-04-21
2 rows in set (0.00 sec)
mysql> select*from foodsystem_waste;
id	item_name	quantity	category	image	date
1	sugar	3	Edible		2026-04-22
2	sugar	3	Inedible		2026-04-22
2 rows in set (0.00 sec)
mysql> select*from foodsystem_donation;
id	ngo_name	food_item	status	quantity
1	karuna	salt	Pending	1
1 row in set (0.00 sec)
mysql> select*from foodsystem_alert;
set (0.19 sec)
foodsystem_report;


Smart Food Waste System
127.0.0.1:8000
Smart Food System
Home
Login
Your Partner in Food Management
OUR SERVICES
Meal Booking
Inventory Tracking
Waste Monitoring
NGO Donation
SYSTEM FEATURES
Dashboard Analytics
Waste Logging
QR Scanner
Reports
Expiry Alerts
Gamification
Type here to search
37°C Mostly sunny
ENG
11:10
24-04-2026
DELL


Smart Food System
127.0.0.1:8000/inventory/
FoodSys
Smart Food Waste Management System
Dashboard
Meal Booking
Inventory
Waste
Donation
QR Scanner
Inventory Management
Item Name
Quantity
dd-mm-yyyy
Add Item
Items
Name
Qty
sugar
30
salt
3
Type here to search
37°C Mostly sunny
ENG
DELL



CONCLUSION

The Smart Food Waste Management System provides an effective and scalable solution to reduce food wastage in institutions. By leveraging digital technologies, it improves efficiency, reduces costs, and promotes sustainable practices.

This project demonstrates how innovation can address real-world problems and contribute to social and environmental well-being.